


# Wireless Power by Resonant Magnetic Coupling

**Chittharanjan S · Deekshitha R · Dani Firas Payyanakadavan**

Department of Electrical Engineering, IIT Madras · EE2025 Engineering Electromagnetics

## 2 Executive Summary

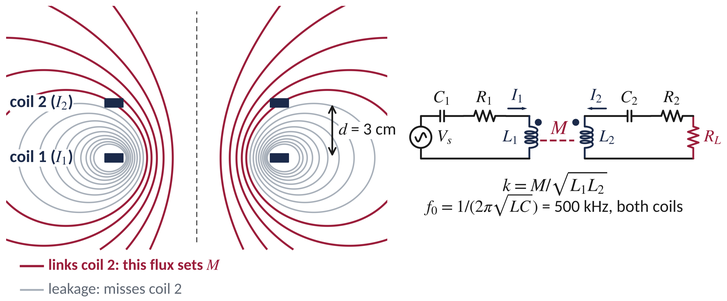
Two identical 10-turn, 10 cm air-core spiral coils, series-tuned to 500 kHz, transfer power across a 1–10 cm gap with no wire between them. At 500 kHz the wavelength is 600 m, so the coils interact through the reactive near field: primary current creates a magnetic field (Ampère), part of its flux links the secondary (mutual inductance $M$), and Faraday's law induces the secondary voltage. Resonance cancels the coil reactance, so performance is set by coupling $k$ and quality factor $Q = 118.3$. A Neumann-integral model gives $k$ falling from 0.51 at 1 cm to 0.030 at 10 cm. A phasor circuit model, confirmed by closed-form theory and LTspice, predicts maximum efficiencies of 97 % at 1 cm, 86 % at 5 cm and 57 % at 10 cm. With the 6 Ω load (optimal at 5 cm), delivered power peaks at critical coupling (6.4 W near 9.75 cm), not where efficiency is highest. All results are analytical and simulated.

## 3 The Physical Concept

**Why it matters.** Wireless charging already powers phones, toothbrushes and medical implants, and is being scaled up for electric vehicles. With no plugs or contacts, a device can be sealed, waterproof, or inside the body. We study the core of it: two identical 10 cm coils, 1 to 10 cm apart, at 500 kHz.

**How it works.** An alternating current in coil 1 makes a magnetic field. Part of that field passes through coil 2 and, because it changes, induces a voltage there (Faraday's law). A capacitor in series with each coil makes it resonate: energy sloshes between the coil's magnetic field and the capacitor's electric field, so a small drive builds a large current. Both coils are tuned to the same $f_0$, so energy passes from one to the other and the load $R_L$ takes it out.

**Why it is short range.** Each coil is 1/6000 of the 600 m wavelength, so it hardly radiates: the field is stored close to the coils. The coupling $k$ falls from 0.51 at 1 cm to 0.03 at 10 cm, and like $1/d^3$ further out.




Left: field lines of coil 1, computed for our coils at $d = 3$ cm; wine lines thread coil 2 and set $M$. Right: the series-series circuit.

## 4 Governing EM Physics

(1) $\nabla \times \boldsymbol{H} = \boldsymbol{J} + \partial \boldsymbol{D}/\partial t \approx \boldsymbol{J} \qquad (k_0 a = \omega a/c \approx 5 \times 10^{-4})$

**Ampère, quasi-static:** the coil is 1/6000 of $\lambda$ = 600 m, so displacement current is negligible.

(2) $\oint \boldsymbol{E}\cdot d\boldsymbol{l} = -d\Phi/dt \;\Rightarrow\; V_2 = -j\omega M I_1$

**Faraday:** the changing shared flux induces the EMF that drives coil 2.

(3) $B_z = \mu_0 N I a^2 / [2(a^2 + z^2)^{3/2}]$

$M = \Phi_{21}/I_1 = (\mu_0/4\pi) \oint\oint d\boldsymbol{l}_1 \cdot d\boldsymbol{l}_2 / r$

**Biot–Savart, Neumann:** flux of coil 1 through coil 2 per amp. Pure geometry; $M_{12} = M_{21}$.

(4) $k = M/\sqrt{(L_1 L_2)}, \qquad k \propto 1/d^3 \;\text{ for }\; d \gg a$

**Coupling:** ours is 0.513, 0.111, 0.030 at 1, 5, 10 cm; $1/d^3$ only for $d \gtrsim 4a$.

(5) $f_0 = 1/(2\pi\sqrt{(LC)}), \qquad Q = \omega_0 L/R, \qquad f_\pm = f_0/\sqrt{(1 \pm k)}$

**Resonance:** 17.13 µH, 5.915 nF → 500 kHz; $Q = 118.3$ at 0.455 Ω. Strong $k$ splits the peak.

(6) $Z_1 I_1 + j\omega M I_2 = V_s, \qquad j\omega M I_1 + Z_2 I_2 = 0$

$Z_{in} = R_1 + \omega_0^2 M^2/(R_2 + R_L) \quad \text{at} \quad f_0$

**Circuit, Poynting:** at $f_0$, $W_m = W_e$ in each coil, so the source sees only resistance.

(7) $\eta_{max} = x\,/\,(1 + \sqrt{(1 + x)})^2, \qquad x = k^2 Q_1 Q_2$

**Efficiency:** at $R_L = R_2\sqrt{(1 + x)}$. At 5 cm $x = 173$, $\eta_{max} = 85.9$ %, $R_L = 6.0$ Ω.

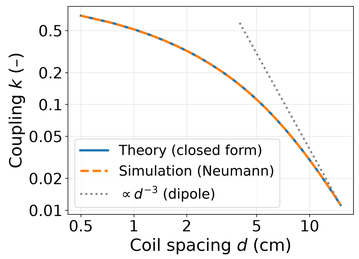
(8) $\delta = \sqrt{(2/(\omega\mu_0\sigma))} = 0.094$ mm (Cu, 500 kHz)

**Skin depth:** current crowds into a skin thinner than the 0.35 mm wire radius, raising $R$.

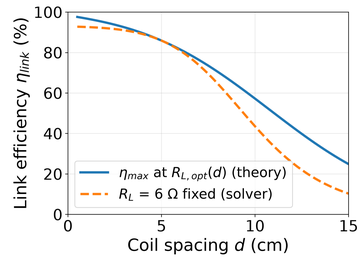
**Boundary conditions:** $\hat{n}\cdot(\boldsymbol{B}_1 - \boldsymbol{B}_2) = 0$, $\hat{n} \times (\boldsymbol{H}_1 - \boldsymbol{H}_2) = \boldsymbol{K}$ Metal forces $B_n \approx 0$ (eddy currents): $L$, $k$ fall and an Al plate 10 mm away adds 87 mΩ. Ferrite guides flux, raising $k$.

**Symbols:** $N$ turns, $a$ coil radius, $d$ spacing; $I_1$, $I_2$ into the dots; $Z_i = R_i + j\omega L_i + 1/(j\omega C_i)$, $R_L$ added in $Z_2$; $\sigma$ conductivity; $\mu_0 = 4\pi \times 10^{-7}$ H/m.

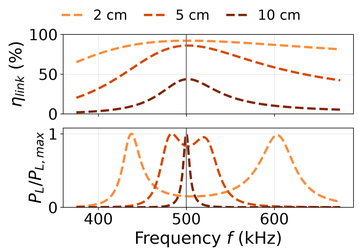
## 5 Analysis and Simulation



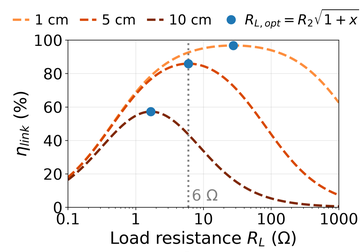

Closed form and Neumann agree to ≤ 0.05 % (0.5–15 cm); $k \propto 1/d^3$ only for $d \gg a$.




$\eta_{max}$ is 97 % at 1 cm, 86 % at 5 cm, 57 % at 10 cm; the fixed 6 Ω load gives 44 % at 10 cm.

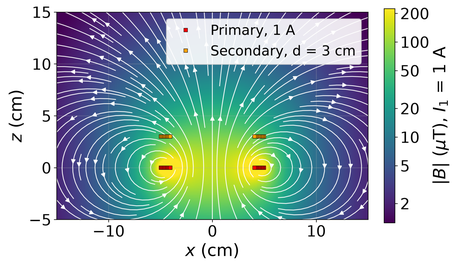



At 2 and 5 cm the power peak splits in two; at 10 cm a single sharp peak sits at $f_0$.




The best load $R_{L,opt} = R_2\sqrt{(1+x)}$ moves with spacing; 6 Ω is best only at 5 cm.

## 5 Analysis and Simulation (cont.)




**Setup**

Planar spiral, $N = 10$
$d_{out} = 100$ mm
wire 0.7 mm, pitch 1.0 mm
$L = 17.13$ µH, $C = 5.915$ nF
$f_0 = 500$ kHz
$R = 0.455$ Ω, $Q = 118.3$
$R_L = 6.0$ Ω
$V_s = 5$ V peak, $R_s = 0$

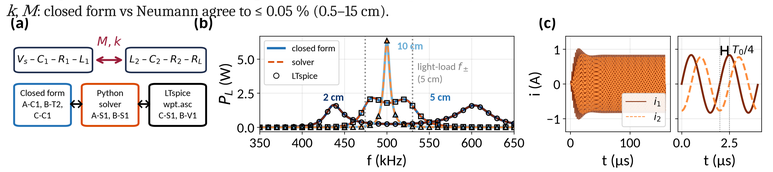
$|B|$ (µT) in the $x$–$z$ plane: primary at 1 A, coil 2 at $d = 3$ cm.

## 6 Demonstration: Model vs Model

- **Validation chain:** closed form ↔ Python Neumann sum + 2 × 2 phasor solver ↔ LTspice sim/wpt.asc. LTspice agrees with the solver to $5 \times 10^{-10}$ % (2, 5, 10 cm, 350–650 kHz).
- Faraday in time: at $f_0$, $V_2 = -j\omega M I_1$ and $Z_2 = R_2 + R_L$ is real, so $i_2$ lags $i_1$ by 90° ($T_0/4 = 0.50$ µs).
- Light-load peaks $f_\pm$ (eq. 5) give $k$ within 0.3 % at 1, 2, 5 cm (5 cm: 474.4 / 530.3 kHz → 0.1109).
- **Finding:** a 6 Ω load pulls them in (5 cm: 483.5 / 519.6 kHz → $k = 0.072$, 35 % low; 10 cm: 1 peak).

| d (cm) | M (µH) | k | ηmax (RL,opt) | η at 6 Ω | PL at 6 Ω (W) | Regime |
|---|---|---|---|---|---|---|
| 1 | 8.79 | 0.513 | 96.8 % (27.6 Ω) | 92.6 % | 0.098 | over-coupled |
| 3 | 3.72 | 0.217 | 92.5 % (11.7 Ω) | 91.0 % | 0.526 | over-coupled |
| 5 | 1.90 | 0.111 | 85.9 % (6.00 Ω) | 85.9 % | 1.79 | over-coupled; design point |
| 10 | 0.51 | 0.030 | 57.2 % (1.67 Ω) | 43.5 % | 6.36 | near critical (k < kc) |

$V_s = 5$ V pk, $R_s = 0$, $R_1 = R_2 = 0.455$ Ω; regime from $k_c = 0.0318$ ($R_L = 6$ Ω, $R_s = 0$).
$k$, $M$: closed form vs Neumann agree to ≤ 0.05 % (0.5–15 cm).




(a) Model chain. (b) $P_L$ (W) vs $f$(kHz) at 2, 5, 10 cm, $R_L = 6$ Ω; markers: LTspice; dotted lines: light-load $f_\pm$ at 5 cm. (c) $i_1$, $i_2$ (A) vs $t$ (µs), 5 cm: $i_2$ lags $i_1$ by 90°. Read $k$ only from light-load peaks.

## 7 Conclusion and References

### Key findings

- $Q = 118.3$: $\eta_{max} = 96.8$ % at 1 cm, 85.9 % at 5 cm ($R_{L,opt} = 6.0$ Ω), 57.2 % at 10 cm.
- Max $\eta \neq$ max power: at 5 cm, $R_{L,opt} = 6.0$ Ω, but $R_L = 79.1$ Ω gives the most power.

### Limitations

- Range: $M \propto 1/d^3$ only for $d \gg a$ (log-log slope −2.1 at 8–10 cm, −2.97 at 50–100 cm).
- AC loss: $R_{dc}$ 0.13 → 0.455 Ω (skin 0.269, prox. 0.136, ESR 0.035, joints 0.015); range 0.38–0.48 Ω.
- Misalignment (3 cm): $\eta = 91.1$ / 88.8 / 84.2 % at $\Delta x = 2$ / 3 / 4 cm; $M = 0$ at 8.22 cm, then $M < 0$.
- Al plate 5 mm behind coil 2: $k$ −56.4 %, $R_2$ +178 mΩ, $L_2$ −51 %; ferrite: $k$ +34.5 %. $\eta$ (3 cm): 92.5 → 24.0 % (63.9 % with $C_2$ retuned).
- Stress ($R_s = 0$): $V_{C1}$ 44.8 / 314 V pk at 5 / 10 cm; $I_1 \leq 11.0$ A pk, 27.5 W (50 Ω source: 0.1 A). Field: 82.3 µT rms at $z = 0$, < 27 µT rms (ICNIRP 2010) beyond 4.7 cm, < 1.84 beyond 15.4 cm.

**Future work:** build and measure a prototype; ferrite backing; adaptive frequency tuning [2].

[1] A. Kurs et al., Science, vol. 317, no. 5834, pp. 83–86, 2007.
[2] A. P. Sample, D. A. Meyer, and J. R. Smith, IEEE Trans. Ind. Electron., vol. 58, no. 2, pp. 544–554, 2011.
[3] S. Y. R. Hui, W. Zhong, and C. K. Lee, IEEE Trans. Power Electron., vol. 29, no. 9, pp. 4500–4511, 2014.
[4] R. K. Shevgaonkar, Electromagnetic Waves. New Delhi: Tata McGraw-Hill, 2006.